Q.No.

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Part-A

1- Apache Hadoop

A- ☆ Apache Hadoop is a tool which is used for data analysis.

☆ It helps us to retrieve & get the data

☆ It helps us to keep data in organised format.

2- Data Serialization

A- ☆ Data Serialization is a process where we keep the data in organised manner

☆ It gives a node to each block

☆ we can indentify the data with the key value

☆ later the data is arranged in ascending or desending order

☆ Data is arranged as per serial number.

Q.No.

3- Hadoop

A- * Hadoop is Java based software which is used for data analysis.

* It helps in various data analytics techniques.

* It is used for data cleaning also.

* It helps us to retrieve & get the data

* It helps us to keep data in organised format.

4- DataNode

A- * Data Node is most important component in HDFS daemons

* It comes in third position after the Name Node & Secondary Node

* It helps us to assign Nodes for the data.

* We can retrieve data in that manner.

Q.No.

5. Machine learning

A. ★ ML is a technique where we provide external training to particular model

★ Based on the training the model provide accurate decisions

★ It also provides results as per the accuracy.

★ It helps in doing repetative tasks & automate the human tasks

★ Types of machine learning techniques

1) Supervised learning

2) Unsupervised learning

3) Semi supervised learning

4) Reinforcement learning.

Q.No.

Part-B

6- HDFS daemons

A. → HDFS - Hadoop distributed file System

* It is a tool which is used for data analysis

* It helps us to retrieve & get the data

* It helps us to keep data in organised format

* It works with different software where we perform different operations

* It also helps in various data analytics techniques

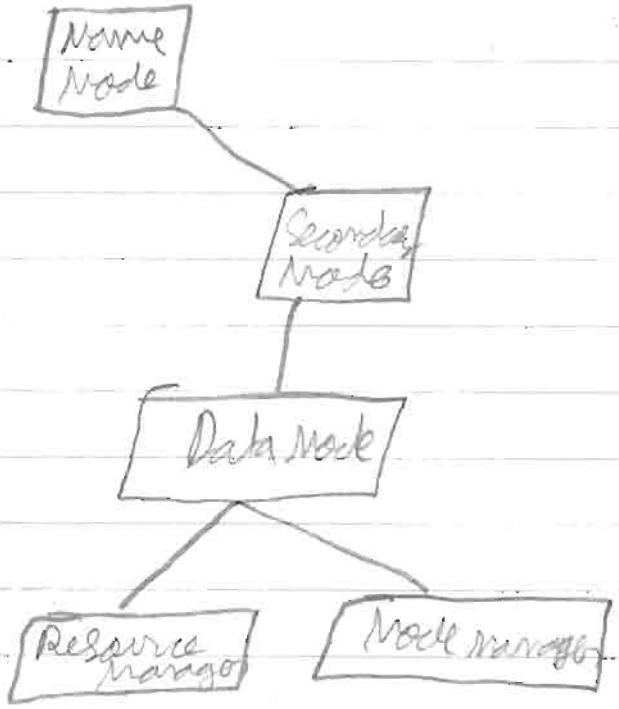
* It also helps in data cleaning, data processing, preprocessing

* There are various components which helps in it.

Q.No.

Various components.

1) Name Node
2) Secondary Node
3) Data Node
4) Resource Manager
5) Node manager.



☆ Name Node

* Name Node is used to assign the Name to the node.

* This one of the main components in HDFS

* It is the first component in the HDFS

* from here only data takes entry

* It responsible for input process

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☆ Secondary Node

☆ It is mainly used to provide backup for Name Node

☆ It is also used to merge ~~to~~ with ~~input~~ Name Node & Support in absence

☆ input format <Key: value>, (Key2, value2)

☆ output format < Key : value>

☆ Data Node

☆ It come in third place after the Name node & Secondary Node

☆ It helps us to assign nodes for the data

☆ We can easily retrieve data in that manner.

☆ Data is retrieved easily with the key value

☆ Data is arranged ~~in~~ ~~a~~ as per the serial

numbers in ascending or descending order..

* Resource Manager

* It provides all the important resources required for tool & software required

* It make necessary arrangements if anything is missing

* It manages with all the resources.

* Node manager

* It manages all the nodes

* It helps in coordination with other components & the nodes

* It is used to maintain data with different nodes

* It helps with memory & data with resources.

Q.No.

8 Machine learning techniques

A- * ML is a technique where we provide external training to particular model

* Based on the training the model provide accurate decisions

* It also provides results as for the accuracy

* It helps in doing repetitive tasks & automate the human tasks

* Types of ML techniques

1) Supervised learning

2) Unsupervised learning

3) Semi Supervised learning

4) Reinforcement learning

Q.No.

☆ Supervised learning

* Here the model is trained with various data sets.

* It takes the data as per the dataset.

* It is trained as per classified data

* It is used ~~during~~ when there are enough datasets

Ex- House price prediction

☆ Unsupervised learning

* Here the model is not trained with dataset

* Here we give new data or raw data for for training purpose

* we might different results for it

Ex - Detection of cyber bullying.

Q.No.

* Semi Supervised learning

* this technique is combination of both
* Here the model is trained with new data & old datasets as well
* It takes both old data & new data
* It can be trained as per the dataset which results in various results
* we might also get different results

ex- loan approval prediction System.

* Reinforcement learning

* It is a special condition for ML
* Here we automate the tasks as per the requirements
* It is used for IOT Purpose also
* ex- Human Scream detection using deep learning.

Q.No.

9- Social media Analytics

A- * Social Media Analytics plays a very important role in day to day life.

* Many People use Social media everyday

* Apps like instagram, facebook etc.

* It tracks our ~~usage~~ usage.

* It controls our activities

* It also collect our data for further usage

* It is used in every Social media platform.

* It reduces cyber bullying in Social media

* Also reduces cyber crimes.

* We can report fake account easily

Q.No.

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Importance

* To track our usage
* To limit overuse
* To reduce cyber bullying
* It reduce cyber crimes
* It helps to report fake accounts

Applications

* Targeted ads - Show only required ads
* Customer aquisition - to aquire more ads
* Feedback - To provide feedback
* product Recommendation

  To recommend the product as per the requirements

* Market analysis

  To analyze different products.

Q.No.

10- Traditional Analytics	Big data Analytics
A ★ It is slow process	★ Fast process
★ Complex System	★ Simple System
★ It uses labours/ manpower for data analysis	★ It uses various tools/ software for data analysis
★ It is used in olden days	★ It is used in modern days
★ lots of paper work	★ No paper work
★ Expensive process	★ cheap process
★ Low initial cost	★ High initial cost
★ Difficult to retrieve data	★ Easy to retrieve data
★ Data can get destroyed	★ Difficult to destroy data
★ Non structured	★ Structured data.
★ Manual process	★ Automatic process

Q.No.

ex- Post office	ex - Data Sieve company

* Traditional Analytics

* It does not uses R programing
* It uses manual process
* More paper work
* labour work

* Big data Analytics

* It uses 'R programming
* along with R it uses various languages like python
* less paper work
* No labour work

Q.No.

R programming

* It is a Scripting language Similar to python

* It is used to data analysis.